

The Lever Generalizes — and It Brakes: A Late, Bidirectional Action-Commitment Lever Across Agent Decisions and Architectures

Toward mechanistic circuit-breakers for irreversible agent actions

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Abstract

A prior result (*The Lever Is Late*) showed that, for a long-horizon coding agent, control of the **finish** decision lives not at the mid-layer “task is done” verdict but in a late, task-matched action-commitment block ~ 30 layers downstream. That left two questions open: is the late lever *specific to termination*, and can it *brake* an action, not just elicit one? We answer both on Qwen3.6-27B over 99 SWE-bench Pro trajectories, using a second, distinct decision available in the same data — commit a file edit (**str_replace_editor**) versus continue reversible exploration (**bash**) — with $n=60$ deterministic decision points per condition, prefill-only patching, and generation-confirmed outcomes. **(1) Generalization (elicit).** At an explore decision, injecting a task-matched *edit*-donor into the late block makes the agent emit a real **edit** call: $0.23 \rightarrow 0.77$ at L59 (position control 0.08, cross-task control 0.48). **(2) The brake (suppress).** At a commit decision, injecting an *explore*-donor collapses the real edit rate $0.48 \rightarrow 0.02$ (96% suppression) at L55, with a same-class control intact (0.55) and the opposite-direction donor boosting to 0.92. **(3) The mechanism is monotonic and bidirectional.** Exact paired McNemar on all 14 per-point conditions yields seven contrasts that survive Holm–Bonferroni ($m=7$, worst $p=7.6e-5$); the discordance structure is the headline: elicit has $c=0$ (the edit-donor only *turns commits on*, never silences) and the brake has $b=0$ (the explore-donor only *turns them off*, never lights). The late lever moves exactly in the donor’s direction with $\sim$ zero off-direction noise. **(4) Cross-architecture.** The late-commitment *geometry* and a donor-specific late *writability* replicate in a different model family (Mistral-7B-Instruct-v0.3): the tool-commit decision is flat/negative through the first half and explodes in the late block, and a task-matched edit-donor raises $P(\text{edit})$ where a same-norm random vector destroys it. We frame the bidirectional late lever as the mechanism for a *mechanistic circuit-breaker*: a single late-layer intervention that blocks an action at its commit point. We demonstrate it on a state-mutating but *undoable* edit; intervening on a genuinely irreversible action (e.g. **send_transaction**) is the named next step. Tool, pre-registration, per-point data, and an adversarial pre-publication evaluation are released.

1 Background and contribution

Recent work [2, 3, 5, 1] established, across five studies on Qwen3.6-27B (a reasoning-tuned 27B model), a detection–control asymmetry for long-horizon agent failure: internal states that *predict* a behavior routinely fail to *control* it. A recent result [1] located control of the **finish** decision in a late action-commitment block (L51–L63), ~ 30 layers downstream of the mid-layer verdict, and only when steered with a *task-matched* donor. We build on a model-agnostic steering tool implementing

three primitives — `LOCATE` (logit-lens per layer), `SWEEP_PATCH` (donor injection per layer), and `STEER_GENERATE` (prefill-only patch of the decision position, then free decoding). This sits in the activation-steering / representation-engineering line [6, 7], applied to a long-horizon agent’s *action channel* across turns rather than a single-turn attribute.

This paper pre-registers and answers two questions that the single **finish** result could not (both pre-registrations are in the supplementary material). First, is the late lever a property of the *action-commitment channel* or an idiosyncrasy of termination? Second, can the lever *suppress* an action at its commit point — the mechanism a safety *circuit-breaker* would need? Our contributions:

1. The late lever **generalizes** from **finish** to a second committal action (commit an edit), by the same locate-patch-steer method, generation-confirmed (§3).
2. The same lever **brakes**: a late explore-donor patch suppresses a real commit by 96% (§4).
3. The effect is **monotonic and bidirectional** — elicit $c=0$, brake $b=0$ — with all seven pre-registered contrasts surviving Holm-Bonferroni on exact paired McNemar (§5).
4. The commitment **geometry and donor-specific writability replicate cross-family** on Mistral-7B (§6).
5. We frame this as a **mechanistic circuit-breaker** and state honestly what is shown (a semi-irreversible proxy) and what is not (a genuinely irreversible action) (§7).

2 Setup

Data and decision points. We reuse the public bundle of 99 SWE-bench Pro [8] trajectories from Qwen3.6-27B (success 40, locked 39, wandering 20). The agent’s action space is three tools: **bash** (3250 calls; reversible exploration), **str_replace_editor** (1101 calls; commits a file edit; semi-irreversible), and **finish** (40). A *commit decision point* is any turn poised between committing an edit and continuing exploration; we reconstruct each faithfully at the tool-choice position, deterministically (sorted traces, ≤ 3 /trajectory), giving $n=60$ edit-decision and $n=60$ bash-decision points. **Patching.** All interventions use a prefill-only patching protocol: the late-position residual is replaced once at the decision token, then the model decodes greedily; patching every step degenerates into repetition and is not used. **Fidelity gate.** Before any causal number, $P(\text{edit})$ at reconstructed edit points (0.474) exceeds that at bash points (0.299), confirming the states reproduce the real commit propensity (decision turns 5.2 vs 4.5 — close, weakening a positional confound; the position-matched control below addresses it directly).

3 The late lever generalizes: eliciting a second action

At a **bash** (explore) decision point we inject a task-matched *edit*-donor late-block residual and ask whether the agent emits a real **str_replace_editor** call (Table 1). The baseline emission rate is 0.23; the task-matched edit-donor raises it to 0.77 at L59. The two controls isolate the effect: a *position* control (an explore-donor at the same points) stays at 0.08 — below baseline, ruling out “any late patch elicits an edit” — and a *cross-task* edit-donor reaches 0.48, well above baseline but below the task-matched donor. As in *The Lever Is Late*, task-matching adds a real increment; unlike **finish**, the **edit** decision also carries a substantial *generic* component (cross-task $\gg$ baseline), so task-specificity is *graded per action*, not absolute.

condition (at bash decision points, $n=60$)	emission rate	exact McNemar vs. baseline
no patch (baseline)	0.23	—
task-matched edit-donor @L59	0.77	$4.7e-10$ ($b=32, c=0$)
task-matched edit-donor @L55	0.62	$2.4e-7$ ($b=23, c=0$)
position control (explore-donor) @L59	0.08	— (vs. donor: $9.1e-13$)
cross-task edit-donor @L59	0.48	— (vs. donor: $7.6e-5$)

Table 1: Elicit. A task-matched late edit-donor makes a stuck-in-exploration agent emit a real edit call.

4 The same lever brakes: suppressing a commit

At an `edit` (commit) decision point we inject an *explore*-donor and measure whether the real edit emission is suppressed (Table 2). The baseline edit rate is 0.48; the explore-donor at L55 collapses it to 0.02 — a 96% suppression. The controls are clean: a same-class (edit-donor) patch at L55 does *not* suppress (0.55), and the opposite intervention — a task-matched edit-donor at L59 — pushes the rate *up* to 0.92. Thus L55 is the brake point and L59 the power point ($0.03 \leftrightarrow 0.92$; Figure 1). A generation-confirmed, single late-layer patch blocks a real commit.

condition (at edit decision points, $n=60$; baseline edit rate 0.48)	edit rate	exact McNemar
explore-donor (brake) @L55	0.02	$7.5e-9$ ($b=0, c=28$)
explore-donor (brake) @L59	0.03	$1.5e-8$ ($b=0, c=27$)
same-class control (edit-donor) @L55	0.55	—
opposite (edit-donor) @L59	0.92	—

Table 2: Brake. A late explore-donor patch suppresses a real commit; same-class control intact.

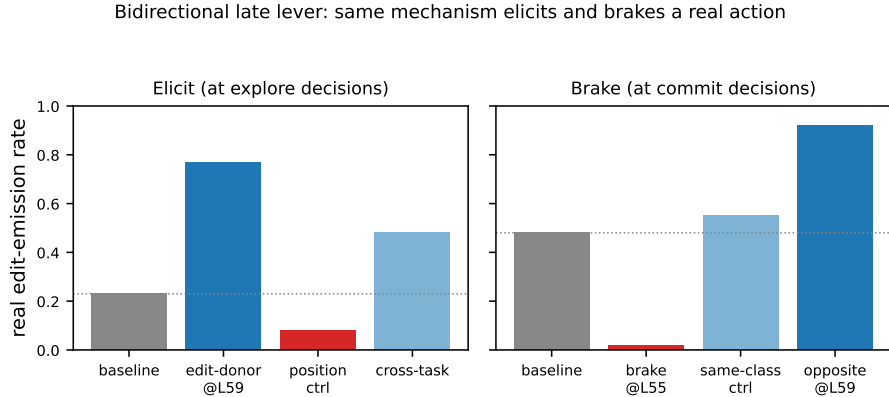


Figure 1: The same task-matched late lever elicits a real edit at an explore decision (left) and brakes a real edit at a commit decision (right), each against position / same-class and cross-task / opposite controls.

5 The mechanism is monotonic and bidirectional

We logged the per-point binary outcome for all 14 conditions and computed exact two-sided paired McNemar [10] tests. All seven pre-registered contrasts survive Holm–Bonferroni [11] ($m=7$; worst, task-specificity, $p=7.6e-5$). The discordance structure is the result: every elicit contrast has $\mathbf{c}=\mathbf{0}$ (the edit-donor turns commits *on*, $b=23-41$, and *never* silences one that already fired), and every brake contrast has $\mathbf{b}=\mathbf{0}$ (the explore-donor turns commits *off*, $c=27-32$, and *never* lights one). The late lever therefore moves the decision *exactly* in the donor’s direction with $\sim$ zero off-direction noise — a structural, not merely a rate-shift, claim. The one-token $\Delta P(\text{edit})$ profile corroborates: monotonic from L23 (-0.003) to L63 ($+0.685$), with mid layers (L23, L31) inert and the explore-donor control going *negative* late (-0.28) — the one-token signature of the brake (Figure 2).

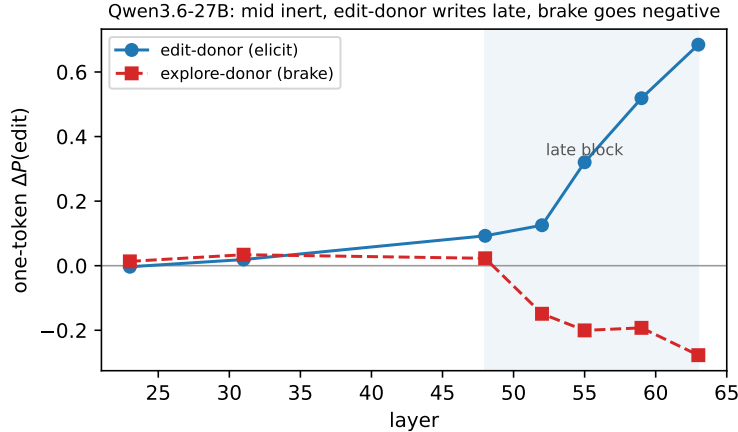


Figure 2: One-token $\Delta P(\text{edit})$ by layer (Qwen3.6-27B). The edit-donor (elicit) is inert mid-stream and rises monotonically to L63; the explore-donor (brake) goes *negative* in the same late block. The lever is late, and bidirectional.

6 Cross-architecture replication

To test whether the late-commitment geometry is architectural rather than Qwen-specific, we ran a lite locate-plus-one-token test on **Mistral-7B-Instruct-v0.3** (a distinct model family) over 30 SWE-bench Pro tasks, with context-defined edit- vs. explore-decision points. **Geometry (robust over three runs)**: the logit-lens edit gap is flat/negative through the first half (L1–L15, ~ -2) and explodes in the late block (L19 $+3.2$, L23 $+6.8$, L27 $+10.3$) — the same mid-flat, late-explosion shape as Qwen. **Writability**: injecting a task-matched edit-context donor into an explore decision raises $P(\text{edit})$ donor-specifically (e.g. $+0.15$ at L27), whereas a same-norm *shuffled* vector *lowers* it (-0.61 at L27); the donor-specific margin is $+0.6-0.76$ in the late block. The donor is inert at L9/L11 and writes from L13 on. Two honest differences from the 27B result: the writable window begins mid-depth ($\sim$ L13, 40%) rather than late-only, and this lite test omits generation confirmation. The qualitative pattern — a late-emerging, direction-specific, written-not-destroyed commitment representation — replicates across families (Figure 3).

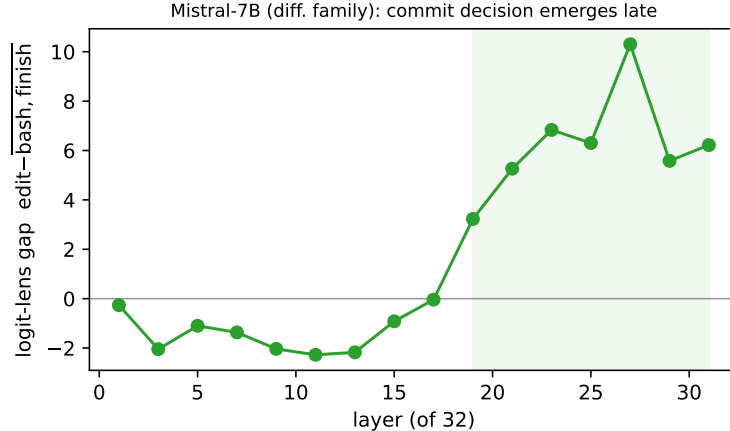


Figure 3: Cross-family LOCATE (Mistral-7B-Instruct-v0.3). The logit-lens commit-vs-explore gap is flat/negative through the first half and explodes in the late block (shaded) — the same late-emergence geometry as Qwen3.6-27B, in a different architecture.

7 Discussion: a mechanistic circuit-breaker, and its honest scope

The bidirectional late lever is the mechanism a safety *circuit-breaker* requires: a single intervention at an action’s commit point that can block it. We emphasize what is and is not shown.

- **Generalization is an existence proof, not a universal law.** It holds for `finish` and one second action, on one model and one task family (plus a cross-family geometry/writability replication). More actions, models, and task families remain.
- **The proxy is semi-irreversible.** `str_replace_editor` mutates state but is undoable. The genuinely irreversible case (`send_transaction`, `delete`) is untested and is the named next step; the circuit-breaker is demonstrated *on a proxy*.
- **The brake redirect is uncontrolled.** Under the brake, `bash` is the dominant action (0.60), but we did not measure the no-brake bash baseline at edit points, so the redirect *magnitude* is not established — we report the brake-state composition, not “routes to exploration”.
- **Emission is tool-token onset.** The elicit rate counts the tool-name prefix in a 16-token greedy continuation; some onsets are malformed, so 0.77 is an upper bound on valid-call elicitation. The brake direction (suppression to 0.02) is robust to this.

These caveats were produced by an adversarial pre-publication evaluation (in the supplementary material) that recomputed every number from the released per-point data (determinism confirmed) and corrected two earlier overclaims; we release it alongside the paper.

Relation to prior work. This extends activation steering / patching from single-turn representations to a long-horizon agent’s action channel, and operationalizes the knowledge–action gap as a *layer* gap with a *bidirectional* late lever — consistent with recent work’s detection≠control thesis and with mechanistic-correctability framings of pausing before irreversible actions.

8 Conclusion

The late action-commitment lever is not specific to termination and is not one-directional: a single task-matched late patch both elicits and brakes a real agent action, monotonically and donor-specifically, with a geometry that replicates across model families. This is the mechanism for a mechanistic circuit-breaker. The next step is to demonstrate it on a genuinely irreversible action and at scale-matched second models.

Artifacts. The model-agnostic steering tool, the two pre-registrations, the per-point data and deterministic decision-point state, the exact-statistics script, and the adversarial pre-publication evaluation are provided in the supplementary material and will be released publicly upon acceptance.

References

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